

SULTAN ZHALIFUNNAS MUSYAFFA

Bekasi, Indonesia | +62 856 9422 9552 | sultanzhalifunnasmusyaffa@gmail.com
linkedin.com/in/sultanzhalifunnasmusyaffa | github.com/SultanZhalifa

PROFESSIONAL SUMMARY

Informatics undergraduate at President University building fullstack web, Android, and AI-integrated systems end to end. Delivered a real-time computer-vision pest-detection platform for an enterprise logistics client using YOLO11 and a Gemini assistant, and shipped a Kotlin Android app backed by 51 automated tests. Comfortable across the stack — React, Next.js, FastAPI, PostgreSQL — with added perspective from cybersecurity and a manufacturing quality background.

EDUCATION

President University — Bachelor of Informatics 2024 – 2027 (Expected)
Cikarang, Indonesia | 5th Semester | Coursework: Algorithms, Data Structures, Database Systems, Software Engineering, Computer Networks

PROJECTS

PestGuard AI — Real-Time Warehouse Pest & Bio-Hazard Detection pestguard-ai.vercel.app

Computer-Vision Platform — PT Kawan Lama Group AI Open Innovation Challenge 2026 (Logistics)

- Engineered a 24/7 computer-vision system detecting snakes, cats, and geckos across up to four camera zones, with sub-second alerts via WebSocket, browser audio, and Telegram.
- Custom-trained a YOLO11-Nano model (~5.2 MB) with CLAHE low-light preprocessing, and added a Gemini 2.0 Flash RAG assistant, role-based access, and PDF/CSV reporting — quantifying an estimated Rp 144–324 million in annual savings per warehouse.
- Tech:** Python, FastAPI, React, YOLO11, OpenCV, Google Gemini, WebSocket, SQLite, Docker.

MiniBookLibrary — Offline-First Android App with 51 Automated Tests github.com/SultanZhalifa/MiniBookLibrary

Personal Project

- Built a Kotlin book-management app (MVVM, Room, Coroutines, Flow) with reading-progress tracking, ISBN auto-fill via the Google Books API, PDF/JSON export, and salted SHA-256 account security.
- Wrote 51 unit tests (JUnit 4, MockK, Turbine) covering authentication, CRUD, ViewModel state, and API parsing — run automatically alongside debug-APK builds through GitHub Actions CI on every push.

DevLog — Developer Progress Tracking Platform devlog-sultanzhalifa.vercel.app

Fullstack Web Application — Personal Project

- Developed a fullstack platform for logging daily learning with a GitHub-style streak heatmap, analytics dashboard, public profiles, and a discovery feed — with GitHub OAuth, a relational follow schema, and unit-tested streak logic behind a CI pipeline.
- Tech:** Next.js 16, TypeScript, Prisma, PostgreSQL (Supabase), Auth.js, Tailwind CSS, Recharts, Vitest, Vercel.

SaringSini — AI Misinformation Checker for Family Chats github.com/SultanZhalifa/SaringSini

Multimodal AI Web App — #JuaraVibeCoding 2026 Competition Entry

- Built a mobile-first PWA that analyzes text, images, and URLs for hoaxes using multimodal Gemini, generating family-friendly response templates across four Indonesian regional languages — hardened with rate limiting, CSP/anti-XSS headers, and a multi-stage Docker build, deployed on Google Cloud Run.
- Tech:** Node.js, Express, Google Gemini, PWA / Service Worker, Docker, Google Cloud Run.

Duitku — Cross-Platform Personal Finance Tracker sultanzhalifa.github.io/Duitku

Flutter Mobile & Web App — Personal Project

- Created an offline-first Flutter app for multi-wallet, multi-currency expense tracking with transfers, recurring automation, budgets, and biometric lock — identical code on Android and web, backed by 17 unit tests. **Tech:** Flutter, Dart, Provider, FL Chart.

WORK EXPERIENCE

PT Honda Prospect Motor Apr 2023 – Apr 2024

Production Worker, Repair AF / Welding Division — Karawang, Indonesia

- Performed daily quality-control checks on welded components, flagging rework that fell outside OEM tolerance standards.

PT Indomobil Trada Nasional (Nissan–Datsun) Mar 2021 – May 2021

Industrial Internship (PRAKERIN), Cikarang — supported workshop inspection and service; earned a "Good" rating from the Workshop Head.

ORGANIZATIONS & ACTIVITIES

HACKSPHERE 2025 — National 48-Hour Hackathon Oct 2025

Event Committee — President University

- Served on the organizing committee for a 48-hour hackathon uniting university students, high-schoolers, and industry professionals in teams of three, supporting participant operations and event logistics throughout.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, Dart, JavaScript, TypeScript, SQL
Web & Mobile: React, Next.js, Node.js, Express, FastAPI, Android (Kotlin, MVVM, Room), Flutter
AI / ML: Google Gemini, LLMs, RAG, Prompt Engineering, YOLO11/YOLOv8, OpenCV, Computer Vision
Testing & QA: JUnit 4, MockK, Turbine, Vitest, GitHub Actions CI, Edge-Case Analysis
Databases: PostgreSQL, MySQL, MongoDB, SQLite, Room ORM, Prisma
Tools & Security: Git, Docker, Linux, REST API, Vulnerability Assessment, Penetration Testing

CERTIFICATIONS

Gemini Certified — Faculty, Student & Educator (3 credentials) Google for Education · 2026
Build an AI Agent · Team Essentials for Designing AI Solutions · Intro to LLMs · Data Literacy IBM SkillsBuild · 2026
C · Java · Haskell Programming · SOLID Principles · Project Management · Career Acceleration with Gemini Dicoding · 2026
Intro to Software Engineering (2026) · Intro to Data Analytics (2024) RevoU
Online Cyber Security Event — Certificate of Appreciation Dibimbing · 2026